

B.Tech. Degree I Semester/II Semester Supplementary Examination

November 2019

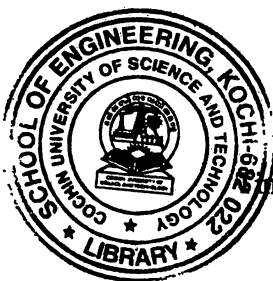
CE/EE/ME/SE GE 15-1105 A & CS/EC/IT GE 15-1205 B

BASIC MECHANICAL ENGINEERING

(2015 Scheme)

Time : 3 Hours

Maximum Marks : 60



PART A

(Answer ALL questions)

(10 × 2 = 20)

- I. (a) Differentiate between open, closed and isolated system.
- (b) List out the classification of steam boiler.
- (c) Define the terms compression ratio, swept volume and clearance volume.
- (d) Define the term COP.
- (e) Define specific speed of a turbine.
- (f) List out the merits and demerits of a solar power plant.
- (g) Define the term depreciation. What are the factors affecting it?
- (h) Compare soldering and brazing.
- (i) Explain wire drawing process.
- (j) Compare flat belt and V belt.

PART B

(4 × 10 = 40)

- II. (a) Name and explain the law that forms the basis of temperature measurement. (2)
- (b) 0.5 kg of air is compressed reversibly and adiabatically from 80 kPa, 60°C to 0.4 MPa, and is expanded at constant pressure to the original volume. Sketch the processes on P-V plane. Compute the heat transfer and work transfer for the whole path? Take $C_p = 1.005 \text{ kJ/kgK}$, $R = 0.287 \text{ kJ/kgK}$ and $\gamma = 1.4$. (8)

OR

- III. (a) Compare boiler mountings and accessories. List out the names. (3)
- (b) What are the different methods of compounding of steam turbines? Explain with neat figures. (7)
- IV. (a) Explain MPFI system in engines. (4)
- (b) Derive an expression for the efficiency of Otto cycle in terms of compression ratio. (6)

OR

- V. (a) Explain the working of a simple vapour compression refrigeration system. (4)
- (b) Explain the working of a four stroke diesel engine with the help of a neat figure. (6)
- VI. (a) Explain the working of a Pelton turbine with the help of a neat figure. (4)
- (b) List out the contributions of Gilbert in industrial engineering. (6)

OR

- VII. (a) Explain the working of a centrifugal pump with the help of a neat figure. (4)
- (b) Explain the working of a thermal power plant with the help of neat figure. (6)
- VIII. (a) Differentiate between hot and cold rolling. (4)
- (b) Explain the principle of annealing hardening and normalizing. (6)

OR

- IX. (a) What are the different types of gears for power transmission? (4)
- (b) Derive an expression for the length of an open belt drive. (6)